

Winkle Lu

Statistical Programmer | Data Visualization & Automation Developer | SAS, R & Python

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Last updated: 12Jul2026

Professional Summary

I am a statistical programmer with 12 years of hands-on clinical trial experience, proficient in CDISC standards and having contributed to 70+ studies across Phases I–III. I led an 18-member programming team for a COVID-19 Phase III trial under tight regulatory timelines, with results published in the New England Journal of Medicine.

Over the past four years, I have deepened my expertise in R, Shiny, Quarto, and Python, focusing on three areas: streamlining TLF and clinical reporting workflows, advancing data visualization, and strengthening data quality through automated validation and anomaly detection. At Junshi, I designed a Python-based automated aCRF tool that extracts variable information from SDTM specifications and generates submission-ready aCRF files, reducing manual work by over 60%. I also developed a metadata-driven clinical data review Shiny application with an interactive patient-level timeline, enabling reviewers to explore domain data without hardcoded configurations — work I presented at R/Pharma 2024 and ShinyConf 2025.

More recently, I expanded into D3.js for richer clinical data presentation. At useR! 2026 (Warsaw), I delivered a Lightning Talk on a modular R and Quarto workflow for reproducible clinical data review, and a Poster on an interactive patient timeline system built with R and D3.js that integrates visits, medications, adverse events, and survival status to support case review and safety signal detection.

Looking ahead, two sessions have been accepted at R/Pharma 2026 APAC: my talk "Upstream Detection of CDISC-Compliant Issues: Raw Data Quality Review with Local LLM and R," and "Interactive QTL/KRI Risk Monitoring with Python," which I am co-presenting.

Performance Summary

- 12 years of clinical trial programming experience with deep expertise in SDTM, ADaM, and regulatory submission.
- Led an 18-member team to complete COVID-19 Phase III programming within 3 months; results published in NEJM.
- Co-led a regulatory-requested RWE study during the COVID-19 program; standardized unstructured hospital RWD through a complete SDTM, ADaM, and TLF pipeline in alignment with regulatory data governance requirements.
- Skilled at cross-functional collaboration, providing medical-monitor support and creating publication-ready output.
- Developed automation tools including an aCRF mapping system and review validation tools, boosting efficiency by 60%.
- Presented at R/Pharma 2024 and ShinyConf 2025, demonstrating advanced R Shiny proficiency.
- Delivered two presentations at useR! 2026 in Warsaw: a Lightning Talk, "Reproducible Clinical Data Review: A Modular R and Quarto Workflow for Transparent Reporting," and a Poster, "Patient Timeline Visualization: Presenting Individual and Population-Level Clinical Trial Journeys with R and D3.js."
- Promoted at three consecutive companies (PAREXEL, Athenex, Junshi), advancing from Programmer to Statistical Programming Manager.

Employment History

Shanghai Junshi Biosciences (Taipei) | Statistical Programming Manager / Principal Statistical Programmer

2021.06 – 2026.04

- Led the programming team for COVID-19 drug development, overseeing 18 programmers to deliver SDTM, ADaM, and TLF outputs within 3 months under tight regulatory timelines; results published in NEJM.
- Co-led a regulatory-requested RWE study; collaborated with hospitals to collect unstructured patient admission and discharge records, standardizing raw RWD through a complete SDTM, ADaM, and TLF pipeline in alignment with regulatory data governance requirements.
- Developed a metadata-driven clinical data review Shiny application adopted for medical monitor support, presented at R/Pharma 2024 and ShinyConf 2025.
- Designed automation tools including an aCRF mapper and outsource QC platform, boosting team efficiency by 60%.
- Provided CDISC training and tool usage workshops to team members.

Athenex (Taipei) | Statistical Programming Manager / Sr. Programmer

2018.09 – 2021.05

- Expanded the programming department at a start-up, more than doubling its project throughput.
- Led SDTM, ADaM, TLFs and submission packages (incl. BIMO, ISS).
- Monitored the quality and correctness of output generated from CROs.
- Reviewed SOPs, improved delivery forms, built company-level macros.
- Mentored junior programmers and conducted regular knowledge sharing.

IQVIA (Taipei) | Statistical Programmer I

2017.01 – 2018.08

- Served as the primary liaison for multiple cross-border collaborations, including partnerships with teams in the United Kingdom, France, and the United States, and consistently delivered each project's objectives on schedule.
- Coordinated project planning, resource allocation, and deliverables.
- Delivered complex SDTM/ADaM datasets and CRF listings.

PAREXEL (Taipei) | Statistical Programmer II / Statistical Programmer I

2014.01 – 2017.01

- Acted as programming lead for international sponsor projects.
- Delivered CDISC-compliant datasets and documentation, including define.xml, reviewer guide.

Technical Skills

SAS	Daily SDTM/ADaM/TLF production
CDISC	SDTM, ADaM, define.xml package
Pinnacle 21 (Community/Enterprise)	Routine validation of SDTM, ADaM, and Define-XML packages; issue resolution and Reviewer's Guide documentation
SQL	Applied via SAS PROC SQL for dataset merging, querying, and data manipulation
R (Tidyverse, Shiny, Quarto)	Production Shiny applications, automated reporting pipelines, and visualization tools
D3.js	Interactive clinical data visualization
Python	Automated tool creation, e.g., aCRF tool
Git / GitHub	Version control for R and Python projects; regular commits, branching, and conflict resolution; public portfolio at github.com/winklelu
Languages	Mandarin (Native), English (Advanced), Taiwanese (Fluent)

Education

- Master of Public Health, Tzu Chi University, 2010
- B.A. in Healthcare Administration, Asia University, 2008

Publications

(earlier publications under Zhi-Sheng Lu)

- Winkle Lu, Upstream Detection of CDISC-Compliant Issues: Raw Data Quality Review with Local LLM and R, R/Pharma 2026 APAC (accepted, forthcoming)
- Yuqi He, Winkle Lu, Interactive QTL/KRI Risk Monitoring with Python, R/Pharma 2026 APAC (accepted, forthcoming)
- Winkle Lu, Reproducible Clinical Data Review: A Modular R and Quarto Workflow for Transparent Reporting (Lightning Talk), useR! 2026
- Winkle Lu, Patient Timeline Visualization: Presenting Individual and Population-Level Clinical Trial Journeys with R and D3.js (Poster), useR! 2026
- Winkle Lu, Reviewing Clinical Data Efficiently with Shiny, ShinyConf 2025
- Winkle Lu, Presenting Clinical Results via CDISC-Compliant Shiny Apps, R/Pharma 2024
- Zhi-Sheng Lu, 2018, Combining quality of productivity and efficiency under highly pressure of lacking time – discussion by view of first-time quality, PharmaSUG – Beijing.
- Shu-Hui Wen, Zhi-Sheng Lu, 2011, Factors affecting the effective number of tests in genetic association studies: A comparative study of three PCA-based methods, Journal of Human Genetics, 56, 428–435 [SCI].